ATMIYA UNIVERSITY

Faculty of Computer Engineering

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title:

AI Business Operating System (AI-BOS)

A Smart, AI & ML Integrated Business Management Platform for Small Businesses

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Certificate

This is to certify that the Software Requirements Specification (SRS) document titled “AI Business Operating System (AI-BOS)” has been prepared by [Your Name], Enrollment No. [Insert Enrollment Number], a student of B.Tech / B.E. Computer Engineering, in partial fulfilment of the requirement for the Final Year Mini Project during the academic year [Insert Year].

The work described in this document has been carried out under my supervision and, to the best of my knowledge, represents an original and satisfactory piece of work suitable for the required academic standard.

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, unambiguous and structured description of the functional and non-functional requirements of the "AI Business Operating System (AI-BOS)" project. This document acts as the primary reference for the design, development, testing and evaluation of the system, developed as a Final Year Mini Project in Computer Engineering. It is intended to create a shared understanding between the student developer, the project guide and the evaluation panel regarding exactly what the system will and will not do.

1.2 Scope

AI-BOS is a web-based business management platform that allows small businesses — such as retail shops, medical stores, educational institutes and restaurants — to manage their day-to-day operations from a single centralised system. Upon registration, a business owner selects a business category, after which the system automatically presents a customised dashboard and navigation menu for that category.

The platform bundles core operational modules (Inventory, Sales, Customers, Suppliers, Employees, Reports, Notifications) with an AI-powered natural-language Business Assistant built on the Google Gemini API and lightweight Machine Learning models built with Scikit-Learn for sales prediction, customer segmentation and product demand classification.

The project is explicitly scoped as a smart business management mini-project and is NOT intended to be a full Enterprise Resource Planning (ERP) system. Enterprise-scale concerns such as multi-branch operations, payroll/taxation compliance, deep learning, and native mobile applications are intentionally excluded from the current scope and listed only under Future Scope (Section 16).

1.3 Objectives

- To design and develop a single web-based platform for managing core business operations — inventory, sales, customers, suppliers and employees.
- To provide a dynamic, category-aware dashboard and navigation experience tailored to Retail, Medical, Education and Restaurant businesses.
- To integrate an AI-powered conversational assistant capable of answering natural-language questions about business data.
- To apply simple, explainable Machine Learning models for sales forecasting, customer segmentation and demand classification.
- To implement secure, role-based access control using JWT authentication.
- To produce a maintainable, well-documented system achievable by a single student within one academic semester.

1.4 Project Vision

The long-term vision of AI-BOS is to demonstrate how modern web technologies, applied machine learning, and generative AI can be combined to give small business owners — who typically cannot afford large ERP systems — an affordable, intelligent, and easy-to-use tool for running their daily operations and making data-driven decisions.

1.5 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
AI-BOS	AI Business Operating System (this project)
JWT	JSON Web Token, used for stateless authentication
API	Application Programming Interface
DRF	Django REST Framework
ML	Machine Learning
AI	Artificial Intelligence
FR	Functional Requirement
NFR	Non-Functional Requirement
BR	Business Rule
CRUD	Create, Read, Update, Delete
RBAC	Role-Based Access Control
ER Diagram	Entity-Relationship Diagram
UI/UX	User Interface / User Experience

1.6 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
- ISO/IEC/IEEE 29148:2018 — Systems and Software Engineering — Requirements Engineering.
- Django REST Framework Official Documentation — https://www.django-rest-framework.org/
- React.js Official Documentation — https://react.dev/
- Scikit-Learn Documentation — https://scikit-learn.org/
- Google Gemini API Documentation — https://ai.google.dev/
- PostgreSQL Official Documentation — https://www.postgresql.org/docs/

2. Overall Description

2.1 Product Perspective

AI-BOS is a new, standalone, self-contained web application. It is not a continuation of any legacy system and does not integrate with any existing enterprise software. It is built as a three-tier system: a React.js single-page frontend, a Django REST Framework backend exposing RESTful APIs, and a PostgreSQL relational database, with an auxiliary Machine Learning service (Scikit-Learn, run within the Django backend) and an external AI service (Google Gemini API) accessed over HTTPS.

2.1.1 System Architecture (High-Level)

```
+-------------------+        HTTPS/REST        +---------------------------+  |  React.js SPA    | <---------------------> |  Django REST
Framework  |  | (Tailwind + MUI)  |       JWT Auth          |        Backend API       |  +-------------------+                    +-----------
--+-------------+                                           |                   +--------------------------+----------+-----------+                |
|                |               +--------v--------+      +---------v---------+  +---------v---------+               |  PostgreSQL DB   |      | ML
Module (Scikit)|  |  Gemini AI Client |              | (Business Data) |      |  Sklearn Models     | |  (External API)  |              +-------
-----------+      +--------------------+  +--------------------+
```

2.2 System Overview

The system exposes a single web application accessed through a browser. After registering their company, an owner selects a Business Category (Retail, Medical, Education or Restaurant). The frontend dynamically renders a dashboard and sidebar menu appropriate to that category, while the backend enforces category-specific and role-specific data access. Core CRUD-based modules cover Inventory, Sales, Customers, Suppliers and Employees. Reports and Notifications operate across all modules. The AI Business Assistant and ML Prediction modules consume the same underlying business data to provide natural-language answers and statistical forecasts, respectively.

2.3 Project Goals

- Deliver a functioning, demonstrable web application within one semester.
- Cover the complete requirement-to-deployment lifecycle for academic evaluation.
- Showcase practical integration of AI (Gemini) and ML (Scikit-Learn) in a real business context.
- Keep the scope realistic for a single-developer academic project while remaining professionally documented.

2.4 Target Users

User Group	Description
Small Business Owners	Primary users who register a company and manage overall operations.
Retail Shops	Businesses selling physical products, tracking stock and daily sales.
Medical Stores	Pharmacies tracking medicine stock, expiry dates and suppliers.
Educational Institutes	Coaching classes / small institutes tracking students, teachers and courses.

User Group	Description
Restaurants	Food businesses tracking orders, customers and daily revenue.
Small Service Businesses	General service-based businesses using the core modules generically.

2.5 User Roles

Role	Access Level	Typical Capabilities
Admin / Owner	Full access	Manage company profile, all modules, employees, reports and settings.
Employee (Staff)	Restricted access	Access only to modules assigned by the Admin (e.g., Sales, Inventory).
System Admin	Platform-level access	Manage registered companies and monitor platform health (Admin Panel).

2.6 Operating Environment

- Client: Any modern web browser (Chrome, Edge, Firefox) on a desktop or laptop, minimum resolution 1280×720.
- Frontend: React.js 18+, Tailwind CSS, Material UI, served as a static single-page application.
- Backend: Python 3.10+, Django 4+, Django REST Framework, deployed on a Linux-based server.
- Database: PostgreSQL 14+.
- External Services: Google Gemini API (internet connectivity required for AI features).

2.7 Design and Implementation Constraints

- The project must be completable by a single Computer Engineering student within one academic semester.
- Only the specified technology stack (React, Django, DRF, PostgreSQL, Scikit-Learn, Gemini API) shall be used.
- The system shall not include Deep Learning, Neural Networks, or LSTM-based models.
- The system shall not include Docker, Kubernetes, CI/CD pipelines, or microservices — a monolithic deployment is sufficient for this scope.
- No native mobile application shall be developed; the web application must be responsive for smaller screens.

2.8 Assumptions

- Users have basic computer literacy and a stable internet connection.
- A valid Google Gemini API key will be available for development and demonstration.
- Business data volume remains at a small-business scale (hundreds to a few thousand records), not enterprise scale.

- The system will primarily be demonstrated with sample/seed data during academic evaluation.

2.9 Dependencies

- Availability and continued free/low-cost access to the Google Gemini API during the development period.
- Availability of open-source libraries: Scikit-Learn, Pandas, NumPy, Django REST Framework.
- PostgreSQL database server availability during development, testing and demonstration.

3. Functional Requirements

This section describes the functional requirements of AI-BOS on a per-module basis. Each requirement is uniquely numbered (FR-001, FR-002, …) in sequential order across the whole document and expressed using the word “shall” to indicate mandatory behaviour.

3.1 Authentication

Purpose

To allow users to securely register, log in, log out, and manage their account credentials.

Description

The Authentication module handles account creation for business owners and employees, credential verification using JWT, session handling, and password management.

Actors

- Business Owner
- Employee
- System

Preconditions

- The user has a valid email address.
- The system is reachable over HTTPS.

Main Flow

- 1. User navigates to the Register or Login screen.
- 2. User submits required credentials.
- 3. System validates input and, on success, issues a JWT access and refresh token pair.
- 4. The user is redirected to the Dashboard.

Alternative Flow

- User selects “Forgot Password” and resets the password via a verification link sent to their registered email.

Exception Flow

- Invalid credentials display an inline error without revealing whether the email or password was incorrect.
- Expired JWT tokens trigger an automatic refresh attempt; failure redirects the user to the Login screen.

Validation Rules

- Email must be in a valid format and unique per account.

- Password must be at least 8 characters and include one number and one special character.
- Phone number, if provided, must be 10 digits.

Business Rules

- A user cannot register twice with the same email.
- JWT access tokens expire after 60 minutes; refresh tokens expire after 7 days.
- Accounts are locked for 15 minutes after 5 consecutive failed login attempts.

Acceptance Criteria

- A registered user can log in and reach their category-specific dashboard within 3 steps.
- Password reset emails are delivered, and the new password is enforced immediately.

Functional Requirements

FR-001: The system shall allow a new user to register using name, email, phone and password.

FR-002: The system shall validate email uniqueness during registration.

FR-003: The system shall issue JWT access and refresh tokens upon successful login.

FR-004: The system shall allow users to log out, invalidating the current refresh token.

FR-005: The system shall allow users to change their password after verifying the current password.

FR-006: The system shall allow users to update basic profile information (name, phone, profile photo).

3.2 Company Registration

Purpose

To capture core company information and set up the category-specific environment for a new business.

Description

During onboarding, the business owner provides company details and selects one business category. This selection drives the dynamic dashboard, sidebar and default data structure for that company.

Actors

- Business Owner

Preconditions

- The user has successfully created and verified an account.

Main Flow

- 1. Owner enters Company Name, Owner Name, Email, Phone and Company Logo.
- 2. The owner selects a Business Category from Retail, Medical, Education or Restaurant.

- 3. The system creates the company profile and initialises the category-specific dashboard and sidebar.
- 4. The owner is redirected to the newly configured Dashboard.

Alternative Flow

- Owner may skip logo upload and add it later from Profile Settings.

Exception Flow

- If the company name already exists for the same owner account, the system prompts for a unique name.
- If the logo upload fails (unsupported format or size), the system retains a default placeholder logo.

Validation Rules

- Company name is required, 3–100 characters.
- Business Category is mandatory and restricted to the four supported categories.
- Company logo, if uploaded, must be PNG/JPG and under 2 MB.

Business Rules

- Each user account maps to exactly one company profile in this project scope (single-tenant per owner).
- Business Category cannot be changed once historical transactional data exists for the company.

Acceptance Criteria

- A new company is registered, and its category-specific dashboard is visible within one workflow, without page reload errors.

Functional Requirements

FR-007: The system shall allow the owner to create a company profile with name, owner name, email, phone and logo.

FR-008: The system shall require selection of exactly one Business Category during registration.

FR-009: The system shall automatically configure the dashboard and sidebar according to the selected category.

FR-010: The system shall prevent changing the Business Category once transactional records exist.

3.3 Dynamic Dashboard

Purpose

To present business-category-relevant key metrics on a single screen immediately after login.

Description

The dashboard renders different widgets depending on the company's Business Category, pulling live summary data from Inventory, Sales, Customers and Suppliers.

Actors

- Business Owner
- Employee

Preconditions

- The user is authenticated and belongs to a registered company.

Main Flow

- 1. User logs in.
- 2. The system detects the company's Business Category.
- 3. The system fetches summary data via Report APIs.
- 4. The system renders the corresponding dashboard layout.

Alternative Flow

- If a widget's data source returns no records yet, the widget displays a friendly empty state instead of an error.

Exception Flow

- If the summary API call times out, the dashboard shows a retry option instead of a blank screen.

Validation Rules

- Dashboard values must refresh on each login and support a manual refresh action.

Business Rules

- Retail dashboards show Products, Sales, Customers and Revenue.
- Medical dashboards show Medicines, low stock, expiry products, and Suppliers.
- Education dashboards show Students, Teachers and Courses.
- Restaurant dashboards show Orders, Customers and Revenue.

Acceptance Criteria

- The correct widget set is displayed for each of the four business categories with accurate, live figures.

Functional Requirements

FR-011: The system shall render a Retail dashboard showing Products, Sales, Customers and Revenue widgets.

FR-012: The system shall render a Medical dashboard showing Medicines, Low Stock, Expiry Products and Suppliers widgets.

FR-013: The system shall render an Education dashboard showing Students, Teachers and Courses widgets.

FR-014: The system shall render a Restaurant dashboard showing Orders, Customers and Revenue widgets.

FR-015: The system shall allow the user to manually refresh dashboard data.

3.4 Dynamic Sidebar

Purpose

To provide category-appropriate navigation so users only see modules relevant to their business type.

Description

The sidebar menu is generated based on the company's Business Category and the logged-in user's role, hiding modules that are not applicable or not permitted.

Actors

- Business Owner
- Employee

Preconditions

- The company's Business Category is already set.

Main Flow

- 1. User logs in.
- 2. System reads the company category and user role.
- 3. System builds and renders the sidebar menu accordingly.

Alternative Flow

- Owners see all modules; employees see only modules explicitly assigned to their role.

Exception Flow

- If the role configuration is missing, the system defaults to showing only the Dashboard and Profile menu items.

Validation Rules

- Menu items must map one-to-one to permitted routes; no link shall lead to an inaccessible page.

Business Rules

- Admin Panel and Employee Management links are visible only to the Admin/Owner role.

Acceptance Criteria

- Navigating the sidebar never exposes a module the user's role or company category does not support.

Functional Requirements

FR-016: The system shall generate the sidebar menu dynamically based on Business Category.

FR-017: The system shall hide or disable sidebar items not permitted for the logged-in user's role.

3.5 Inventory Management

Purpose

To let businesses track products, categories and stock levels accurately.

Description

The Inventory module supports adding, editing, deleting, searching and viewing products, along with category tagging and stock quantity tracking that feeds the Sales and Notification modules.

Actors

- Business Owner
- Employee (with Inventory access)

Preconditions

- The user is authenticated and has Inventory module permission.

Main Flow

- 1. User opens Inventory module.
- 2. User adds a product with name, category, price, and stock quantity.
- 3. System validates and saves the product.
- 4. Product appears in the inventory list and becomes available for Sales.

Alternative Flow

- User searches or filters products by name, category or stock status instead of browsing the full list.

Exception Flow

- Duplicate product names within the same company prompt a confirmation dialog before saving.
- Attempting to delete a product referenced in existing invoices is blocked with an explanatory message.

Validation Rules

- Product name is required, 2–100 characters.
- Price must be a positive number.

- Stock quantity must be a non-negative integer.

Business Rules

- Products with stock quantity below a configurable threshold are automatically flagged as Low Stock.
- Deleted products are soft-deleted (marked inactive) to preserve historical invoice integrity.

Acceptance Criteria

- A product can be added, edited, searched and removed, with stock quantity correctly reflected after each Sales transaction.

Functional Requirements

FR-018: The system shall allow authorized users to add a new product with name, category, price and stock quantity.

FR-019: The system shall allow editing of existing product details.

FR-020: The system shall allow soft-deletion of a product not referenced in existing invoices.

FR-021: The system shall allow searching and filtering products by name, category and stock status.

FR-022: The system shall automatically decrement stock quantity when a sale is recorded.

FR-023: The system shall flag a product as Low Stock when quantity falls below the configured threshold.

3.6 Sales Module

Purpose

To record sales transactions and generate invoices for customers.

Description

The Sales module allows staff to create invoices by selecting a customer and one or more products, automatically calculating totals and updating inventory and customer history.

Actors

- Business Owner
- Employee (with Sales access)

Preconditions

- At least one product and one customer exist (or a walk-in customer option is used).

Main Flow

- 1. User creates a new invoice.
- 2. User selects a customer and adds one or more products with quantities.
- 3. System calculates line totals, tax (if applicable) and grand total.

- 4. User confirms the invoice.
- 5. System saves the invoice, decrements stock, and updates customer purchase history.

Alternative Flow

- User reprints or exports a previously generated invoice from Invoice History.

Exception Flow

- If a selected product's stock is insufficient for the requested quantity, the system blocks invoice confirmation and shows the available quantity.

Validation Rules

- At least one product line is required per invoice.
- Quantity per line must not exceed available stock.

Business Rules

- Every confirmed invoice is immutable; corrections require a separate credit note (out of scope) or a new invoice.

Acceptance Criteria

- An invoice can be created end-to-end, correctly updates stock and customer history, and appears in Invoice History and Reports.

Functional Requirements

FR-024: The system shall allow creation of an invoice with one or more product line items.

FR-025: The system shall automatically calculate line totals and the invoice grand total.

FR-026: The system shall prevent confirming an invoice when requested quantity exceeds available stock.

FR-027: The system shall maintain a searchable Invoice History for each company.

FR-028: The system shall associate each invoice with a customer record.

3.7 Customer Module

Purpose

To maintain a record of customers and their purchase history for service and analytics purposes.

Description

The Customer module supports standard CRUD operations on customer profiles and displays each customer's transaction history, which also feeds the AI Assistant and Customer Segmentation ML feature.

Actors

- Business Owner
- Employee

Preconditions

- The user has Customer module access.

Main Flow

- 1. User adds a customer with name, phone and optional email.
- 2. Customer becomes available for selection during Sales.
- 3. System accumulates purchase history against the customer record automatically.

Alternative Flow

- User updates customer contact details after a change of information.

Exception Flow

- Attempting to delete a customer with existing invoice history prompts a warning and requires confirmation before soft-deletion.

Validation Rules

- Customer name is required; phone number must be a valid 10-digit number if provided.

Business Rules

- Duplicate customers are identified by matching phone number within the same company.

Acceptance Criteria

- A customer can be added, updated, and their full purchase history reviewed accurately.

Functional Requirements

FR-029: The system shall allow adding a new customer with name and contact details.

FR-030: The system shall allow updating and soft-deleting customer records.

FR-031: The system shall display purchase history per customer.

3.8 Supplier Module

Purpose

To track supplier information for procurement and stock replenishment reference.

Description

The Supplier module stores supplier contact and product-category details, supporting the Medical and Retail dashboards' supplier widgets.

Actors

- Business Owner
- Employee

Preconditions

- The user has Supplier module access.

Main Flow

- 1. User adds a supplier with name, contact details and supplied product category.
- 2. Supplier appears in the supplier list for reference during restocking.

Alternative Flow

- User updates supplier details when contact information changes.

Exception Flow

- Deleting a supplier linked to recent restocking records requires confirmation.

Validation Rules

- Supplier name and phone number are required fields.

Business Rules

- Supplier records are company-scoped and not shared across companies.

Acceptance Criteria

- A supplier can be added, edited, viewed and removed without affecting unrelated modules.

Functional Requirements

FR-032: The system shall allow adding a new supplier with name, contact and product category.

FR-033: The system shall allow editing and soft-deleting supplier records.

FR-034: The system shall list all suppliers with search capability.

3.9 Employee Module

Purpose

To manage staff records and control their access to system modules.

Description

The Employee module allows the Owner to add staff accounts, assign role-based module permissions, and maintain a directory of employees.

Actors

- Business Owner

Preconditions

- The logged-in user has the Owner/Admin role.

Main Flow

- 1. Owner adds an employee with name, email, phone and assigned module permissions.
- 2. System creates a linked login account for the employee.

- 3. Employee logs in and sees only permitted modules via the Dynamic Sidebar.

Alternative Flow

- Owner updates an employee's assigned permissions at any time.

Exception Flow

- Attempting to delete an employee with active session data logs them out and revokes access immediately.

Validation Rules

- Employee email must be unique within the company.

Business Rules

- Only the Owner/Admin role can create, edit or remove employee accounts.

Acceptance Criteria

- An employee account can be created with specific permissions, and the assigned employee's sidebar reflects exactly those permissions.

Functional Requirements

FR-035: The system shall allow the Owner to add an employee account with assigned module permissions.

FR-036: The system shall allow the Owner to update or revoke an employee's permissions.

FR-037: The system shall list all employees with their assigned roles.

3.10 Reports

Purpose

To provide summarized, visual insight into Sales, Inventory and Customer data.

Description

The Reports module aggregates transactional data into Sales Reports, Inventory Reports and Customer Reports, presented with charts for a selected date range.

Actors

- Business Owner
- Employee (with Reports access)

Preconditions

- Sufficient transactional data exists for the selected period.

Main Flow

- 1. User selects a report type and date range.
- 2. System aggregates the relevant data.

- 3. System renders summary figures and charts.

Alternative Flow

- User exports report data to CSV for offline use.

Exception Flow

- If no data exists for the selected range, the system displays an informative empty state rather than an empty chart.

Validation Rules

- Date range End Date must not be earlier than Start Date.

Business Rules

- Reports are always scoped to the logged-in user's company; cross-company data is never shown.

Acceptance Criteria

- Reports render accurate totals matching underlying Sales, Inventory and Customer records for a given date range.

Functional Requirements

FR-038: The system shall generate a Sales Report for a selected date range with totals and charts.

FR-039: The system shall generate an Inventory Report showing stock levels and low-stock items.

FR-040: The system shall generate a Customer Report showing top customers by purchase value.

FR-041: The system shall allow exporting report data to CSV.

3.11 Notification System

Purpose

To proactively alert users about important business events.

Description

The Notification module generates in-app alerts for low stock conditions and successful invoice creation, keeping the Owner informed without manual checking.

Actors

- System (automatic trigger)
- Business Owner
- Employee

Preconditions

- The relevant module (Inventory or Sales) has generated a qualifying event.

Main Flow

- 1. A product's stock falls below threshold, or an invoice is created.
- 2. System generates a notification record.
- 3. User sees the notification in the notification panel/bell icon.

Alternative Flow

- User marks a notification as read or clears all notifications.

Exception Flow

- If notification delivery fails, the event is still logged and visible on next login.

Validation Rules

- Notifications must reference a valid source record (product or invoice).

Business Rules

- Notifications are retained for 30 days before automatic archival.

Acceptance Criteria

- A Low Stock event and an Invoice Created event each reliably produce a corresponding, visible notification.

Functional Requirements

FR-042: The system shall generate a notification when a product's stock falls below the configured threshold.

FR-043: The system shall generate a notification when a new invoice is created.

FR-044: The system shall allow users to mark notifications as read.

3.12 Role Based Access

Purpose

To ensure users can only access data and actions appropriate to their assigned role.

Description

RBAC is enforced at both the frontend (UI visibility) and backend (API-level permission checks) layers, based on the Owner/Admin and Employee roles defined in Section 2.5.

Actors

- Business Owner
- Employee
- System

Preconditions

- A user is authenticated and has an assigned role.

Main Flow

- 1. User attempts an action.
- 2. System checks the user's role and permissions against the requested API endpoint.
- 3. System allows or denies the action accordingly.

Alternative Flow

- N/A — access control is enforced uniformly.

Exception Flow

- Any unauthorized API request returns an HTTP 403 Forbidden response with no data leakage.

Validation Rules

- Every protected API endpoint must verify both authentication (valid JWT) and authorization (role permission).

Business Rules

- Backend authorization checks are authoritative; frontend hiding of menu items is a UX convenience only, not a security boundary.

Acceptance Criteria

- An Employee without Inventory permission cannot access Inventory data even via direct API calls.

Functional Requirements

FR-045: The system shall enforce role-based authorization on every protected API endpoint.

FR-046: The system shall return HTTP 403 for unauthorized access attempts without exposing internal data.

3.13 AI Business Chat Assistant

Purpose

To let business owners query their business data using natural language instead of manually navigating reports.

Description

The AI Assistant uses the Google Gemini API to interpret natural-language questions, retrieve relevant business data from the database, and return a clear natural-language response. No AI model training is performed; Gemini is used purely for language understanding and generation.

Actors

- Business Owner
- Employee (with AI Assistant access)

Preconditions

- A valid Gemini API key is configured.
- The company has sufficient data to answer typical queries.

Main Flow

- 1. User types a question into the AI Assistant chat box (e.g., “Which products have low stock?”).
- 2. Backend retrieves relevant structured data from the database based on the query intent.
- 3. Backend sends the question plus retrieved context to the Gemini API.
- 4. Gemini returns a natural-language answer, which is displayed in the chat window.

Alternative Flow

- User asks a follow-up question within the same chat session, retaining prior context.

Exception Flow

- If the Gemini API is unreachable, the assistant displays a graceful fallback message instead of failing silently.
- If a query cannot be mapped to available data, the assistant explains it cannot answer rather than guessing.

Validation Rules

- Only company-scoped data may be sent as context to the Gemini API — never another company's data.

Business Rules

- The assistant answers strictly from real business data; it does not fabricate figures.

Acceptance Criteria

- Sample queries — “Which products have low stock?”, “Show today's sales summary”, “Who is my best customer?”, “Generate inventory summary”, “Explain today's sales” — each return an accurate, relevant natural-language response.

Functional Requirements

FR-047: The system shall provide a chat interface for natural-language business queries.

FR-048: The system shall retrieve relevant company-scoped data before calling the Gemini API.

FR-049: The system shall display the Gemini-generated natural-language response to the user.

FR-050: The system shall handle Gemini API failures gracefully with a fallback message.

3.14 Machine Learning Prediction

Purpose

To provide simple, explainable predictive insights using historical business data.

Description

Three lightweight ML features are provided using Scikit-Learn: Sales Prediction (Linear Regression), Customer Segmentation (K-Means Clustering) and Product Demand Prediction (Decision Tree). No Deep Learning, Neural Networks or LSTM models are used.

Actors

- Business Owner
- System (scheduled/on-demand prediction)

Preconditions

- Sufficient historical data exists (minimum sample size per model, see Section 8).

Main Flow

- 1. User opens the Predictions section or dashboard widget.
- 2. Backend loads historical data and the relevant trained model.
- 3. Backend computes and returns the prediction.
- 4. Frontend displays the prediction with a simple explanation.

Alternative Flow

- User manually triggers re-training/refresh of a model after new data is added.

Exception Flow

- If insufficient historical data exists, the system informs the user rather than returning an unreliable prediction.

Validation Rules

- Predictions are computed only from the requesting company's own historical data.

Business Rules

- Predictions are advisory only and clearly labelled as estimates, not guarantees.

Acceptance Criteria

- Sales Prediction returns a next-day estimate; Customer Segmentation labels customers as VIP, Regular or New; Demand Prediction labels products as High, Medium or Low demand — each grounded in the company's actual data.

Functional Requirements

FR-051: The system shall predict next day's sales using a Linear Regression model on historical sales data.

FR-052: The system shall segment customers into VIP, Regular and New using K-Means Clustering.

FR-053: The system shall classify product demand as High, Medium or Low using a Decision Tree model.

FR-054: The system shall clearly label all ML outputs as estimates.

3.15 Admin Panel

Purpose

To give a platform-level administrator visibility and control over registered companies.

Description

The Admin Panel is a restricted section allowing a System Admin to view registered companies, monitor basic platform usage, and deactivate a company account if required.

Actors

- System Admin

Preconditions

- The logged-in user holds the System Admin role.

Main Flow

- 1. Admin logs into the Admin Panel.
- 2. Admin views the list of registered companies and summary statistics.
- 3. Admin may deactivate or reactivate a company account.

Alternative Flow

- N/A

Exception Flow

- Non-Admin users attempting to access Admin Panel routes are redirected with an access-denied message.

Validation Rules

- Only the System Admin role can access this module.

Business Rules

- Company account status changes take effect immediately across the platform.

Acceptance Criteria

- The Admin Panel accurately lists all registered companies and their active/inactive status.

Functional Requirements

FR-055: The system shall provide a System Admin view listing all registered companies.

FR-056: The system shall allow the System Admin to activate or deactivate a company account.

3.16 Profile Management

Purpose

To allow users to view and update their personal and company profile information.

Description

The Profile module lets a user update personal details, change their password, and (for Owners) update company details such as logo and contact information.

Actors

- Business Owner
- Employee

Preconditions

- The user is authenticated.

Main Flow

- 1. User opens Profile.
- 2. User edits editable fields (name, phone, photo, and for Owners, company details).
- 3. System validates and saves changes.

Alternative Flow

- N/A

Exception Flow

- Uploading an unsupported file format for a profile photo is rejected with a clear error message.

Validation Rules

- Email address is read-only after registration to preserve login integrity.

Business Rules

- Only the Owner can edit company-level profile fields; Employees can edit only their personal fields.

Acceptance Criteria

- Profile changes are saved and reflected immediately across the application.

Functional Requirements

FR-057: The system shall allow users to update their personal profile information.

FR-058: The system shall allow the Owner to update company profile information.

FR-059: The system shall restrict email address changes after account creation.

4. Non-Functional Requirements

4.1 Performance

NFR-001: The system shall load the Dashboard within 3 seconds under normal load (single-user demonstration environment).

NFR-002: API responses for standard CRUD operations shall complete within 1 second for datasets up to 5,000 records.

NFR-003: The AI Assistant shall return a response within 5 seconds under normal Gemini API latency.

4.2 Security

NFR-004: All client-server communication shall occur over HTTPS.

NFR-005: Passwords shall be stored using a salted hashing algorithm (Django's default PBKDF2 or stronger).

NFR-006: JWT tokens shall be used for stateless authentication with defined expiry as per Section 3.1.

4.3 Availability

NFR-007: The system shall target 99% uptime during the academic evaluation and demonstration period.

4.4 Maintainability

NFR-008: The codebase shall follow a modular structure (separate Django apps per module) to simplify future maintenance and extension.

4.5 Reliability

NFR-009: The system shall handle invalid input gracefully without crashing, returning descriptive error messages.

4.6 Scalability

NFR-010: The database schema shall support horizontal growth in company count and record volume without structural redesign, within small-business data scale.

4.7 Usability

NFR-011: The user interface shall follow consistent design patterns using Material UI components across all screens.

NFR-012: A first-time user shall be able to complete company registration and reach the dashboard without external assistance.

4.8 Accessibility

NFR-013: The interface shall maintain a minimum colour contrast ratio suitable for readability and use legible font sizing across screens.

4.9 Compatibility

NFR-014: The application shall function correctly on the latest two versions of Chrome, Edge and Firefox.

4.10 Portability

NFR-015: The backend shall be deployable on any standard Linux server supporting Python 3.10+ and PostgreSQL 14+.

4.11 Logging

NFR-016: The system shall log authentication events, and Admin actions for audit purposes (see Section 10).

4.12 Backup and Recovery

NFR-017: The database shall be backed up on a scheduled basis (e.g., daily) in the deployment environment.

NFR-018: A documented recovery procedure shall allow restoration of the latest backup within a reasonable timeframe.

5. User Interface Requirements

All screens follow a consistent Material UI design language with Tailwind CSS utility styling, a persistent top navigation bar, and a category-aware collapsible sidebar. Every screen is responsive and adapts to tablet and mobile viewport widths using a fluid grid layout; on narrow screens the sidebar collapses into a toggled drawer menu.

Screen	Purpose	Key Fields	Key Actions	Validation	Navigation
Home / Landing	Introduce AI-BOS and direct visitors to Login or Register.	—	Login, Register, Learn More	N/A	Links to Login and Register screens.
Login	Authenticate an existing user.	Email, Password	Login, Forgot Password	Valid email format; password required	On success, navigates to Dashboard.
Register	Create a new user account.	Name, Email, Phone, Password, Confirm Password	Register, Back to Login	All fields required; password match check	On success, navigates to Company Registration.
Company Registration	Capture company details and business category.	Company Name, Owner Name, Email, Phone, Logo Upload, Business Category (select)	Save & Continue	Company name and category required	Navigates to Dashboard on success.
Dashboard	Show category-specific key business metrics.	— (read-only widgets)	Refresh, Quick Actions	N/A	Sidebar links to all permitted modules.
Inventory	List, add, edit and search products.	Product Name, Category, Price, Stock Qty, Search Bar	Add Product, Edit, Delete, Search	Name and price required; price > 0	Row-level Edit/Delete open modals; list is paginated.
Sales	Create invoices and view sales history.	Customer Select, Product Select, Quantity, Auto Totals	Create Invoice, View History	At least one line item; quantity ≤ stock	Confirm navigates to Invoice History.
Customers	Manage customer records and view history.	Name, Phone, Email, Search Bar	Add Customer, Edit, Delete, View History	Name required; phone format check	Customer row opens purchase history view.
Suppliers	Manage supplier records.	Name, Phone, Product Category, Search Bar	Add Supplier, Edit, Delete	Name and phone required	Standard list-detail navigation.

Screen	Purpose	Key Fields	Key Actions	Validation	Navigation
Employees	Manage staff accounts and permissions.	Name, Email, Phone, Module Permission Checkboxes	Add Employee, Edit, Deactivate	Email uniqueness check	Owner-only screen; hidden for Employees.
Reports	View and export Sales, Inventory and Customer reports.	Report Type Select, Date Range Picker	Generate, Export CSV	End date ≥ start date	Charts render inline; export triggers download.
AI Assistant	Chat interface for natural-language business queries.	Chat Input Box	Send	Non-empty message required	Scrollable chat history within session.
Predictions	View ML-based Sales, Segmentation and Demand insights.	— (read-only, category selectable)	Refresh Prediction	N/A	Accessible from Dashboard quick links.
Profile	View and edit personal (and company, for Owner) details.	Name, Phone, Photo, Company Fields (Owner only)	Save Changes, Change Password	Standard field validation	Accessible from top navigation avatar menu.
Settings	Configure notification preferences and low-stock threshold.	Low Stock Threshold, Notification Toggles	Save Settings	Threshold must be a non-negative integer	Accessible from Profile menu.
Admin Panel	System Admin view of all registered companies.	Company List, Status Filter	Activate, Deactivate	N/A	Restricted to System Admin role only.

5.1 Responsive Design Guidelines

- Desktop (≥ 1200px): full sidebar, multi-column dashboard widgets, data tables with all columns visible.
- Tablet (768–1199px): collapsible sidebar, dashboard widgets reflow to two columns.
- Mobile (< 768px): sidebar becomes a drawer menu, tables convert to stacked card views, single-column dashboard widgets.

6. Database Requirements

AI-BOS uses a relational PostgreSQL database. The schema is company-scoped: nearly every business table carries a company foreign key so that data from different companies remains logically isolated within a shared database (single database, multi-tenant-by-row design), which is appropriate for this project's scale.

6.1 Major Tables

Table	Description	Key Relationships
Users	Login accounts for Owners, Employees and System Admins.	One User → One Company (for Owners); referenced by Employees.
Companies	Registered business profiles.	One Company → Many Products, Customers, Suppliers, Employees.
BusinessCategories	Fixed lookup: Retail, Medical, Education, Restaurant.	One Category → Many Companies.
Products	Inventory items with price and stock quantity.	Many Products → One Company; referenced by InvoiceItems.
Customers	Customer contact and history reference.	One Customer → Many Invoices.
Suppliers	Supplier contact and category information.	Many Suppliers → One Company.
Employees	Staff records linked to Users and permission sets.	One Employee → One User; One Company → Many Employees.
Invoices	Sales transaction headers.	One Invoice → One Customer; One Invoice → Many InvoiceItems.
InvoiceItems	Line items per invoice.	Many InvoiceItems → One Invoice; Many InvoiceItems → One Product.
Notifications	System-generated alerts.	Many Notifications → One Company.
Reports	Cached/generated report metadata.	Many Reports → One Company.
Predictions	Stored ML prediction outputs.	Many Predictions → One Company.
AILogs	Log of AI Assistant queries and responses (for audit and improvement).	Many AILogs → One Company; Many AILogs → One User.

6.2 Entity-Relationship Overview (High-Level)

Companies (1)---(M) Products (1)---(M) InvoiceItems (M)---(1) Invoices (M)---(1) Customers | |---(M) Suppliers |---(M) Employees ---(1) Users |---(M) Notifications |---(M) Reports |---(M) Predictions |---(M) AILogs ---(1) Users

6.3 Data Integrity Rules

- Every business table (Products, Customers, Suppliers, Employees, Invoices, Notifications, Reports, Predictions, AILogs) enforces a NOT NULL foreign key to Companies.
- Deletions of referenced records (e.g., Products used in Invoices) use soft-delete (an is_active flag) rather than hard deletion, to preserve historical integrity.

- Invoices and InvoiceItems are never edited after confirmation; corrections require a new record.

7. API Requirements

The backend exposes a RESTful API built with Django REST Framework, secured using JWT Bearer tokens. All endpoints are prefixed with /api/v1/ and return JSON. Below is a representative, non-exhaustive summary of endpoint groups.

API Group	Representative Endpoints	Methods
Authentication APIs	/auth/register, /auth/login, /auth/refresh, /auth/logout	POST
Company APIs	/companies, /companies/{id}	GET, POST, PUT
Inventory APIs	/products, /products/{id}	GET, POST, PUT, DELETE
Sales APIs	/invoices, /invoices/{id}	GET, POST
Customer APIs	/customers, /customers/{id}, /customers/{id}/history	GET, POST, PUT, DELETE
Supplier APIs	/suppliers, /suppliers/{id}	GET, POST, PUT, DELETE
Employee APIs	/employees, /employees/{id}, /employees/{id}/permissions	GET, POST, PUT, DELETE
Report APIs	/reports/sales, /reports/inventory, /reports/customers	GET
AI APIs	/ai/chat	POST
Prediction APIs	/predictions/sales, /predictions/segmentation, /predictions/demand	GET
Admin APIs	/admin/companies, /admin/companies/{id}/status	GET, PUT

7.1 API Conventions

- All protected endpoints require a valid JWT Bearer token in the Authorization header.
- List endpoints support pagination via ?page and ?page_size query parameters.
- Errors follow a consistent JSON structure: { "error": { "code": ..., "message": ... } }.
- All write endpoints validate payloads server-side, independent of frontend validation.

8. Machine Learning Requirements

AI-BOS uses three simple, explainable Scikit-Learn models. No Deep Learning, Neural Network or LSTM techniques are used, consistent with the project's academic mini-project scope.

8.1 Sales Prediction — Linear Regression

Aspect	Detail
Dataset	Historical daily sales totals for the company (from Invoices), minimum 30 days recommended.
Input Features	Date-derived features (day of week, day of month), rolling average of past N days' sales.
Output	Predicted sales total for the next day.
Training	Model retrained periodically (e.g., on-demand or nightly) using scikit-learn LinearRegression on the company's own historical data.
Prediction Flow	Retrieve recent sales → build feature vector → model.predict() → return estimate with a confidence note.
Evaluation	Mean Absolute Error (MAE) and R^2 score computed on a held-out validation split.
Model Storage	Serialized per-company model file (pickle/joblib) stored server-side, reloaded for prediction requests.

8.2 Customer Segmentation — K-Means Clustering

Aspect	Detail
Dataset	Aggregated per-customer purchase frequency and total spend from Invoices.
Input Features	Purchase frequency, total spend, recency of last purchase.
Output	Cluster label mapped to VIP, Regular or New.
Training	K-Means with k=3 fitted on the company's customer feature set.
Prediction Flow	Compute features for all customers → assign clusters → map clusters to labels by spend ranking.
Evaluation	Qualitative validation against expected spend ordering; silhouette score used as a sanity check.
Model Storage	Cluster centers stored per company; recomputed as new invoice data accumulates.

8.3 Product Demand Prediction — Decision Tree

Aspect	Detail
Dataset	Historical per-product sales volume and stock turnover from Invoices and Products.
Input Features	Average units sold per week, current stock level, product category.
Output	Demand classification: High, Medium or Low.
Training	scikit-learn DecisionTreeClassifier trained on labelled historical demand buckets.

Aspect	Detail
Prediction Flow	Compute recent features per product → model.predict() → label displayed alongside the product in Inventory/Predictions screens.
Evaluation	Accuracy and confusion matrix on a held-out validation split.
Model Storage	Serialized per-company model file, reloaded for prediction requests.

8.4 General ML Constraints

- All models are trained and evaluated per company on that company's own data — no cross-company data mixing.
- Predictions are recomputed on demand or on a scheduled basis; the project does not require real-time streaming ML.
- Deep Learning, Neural Networks and LSTM are explicitly out of scope for this project.

9. Artificial Intelligence Requirements

9.1 Gemini API Integration

The AI Business Assistant integrates with the Google Gemini API over HTTPS using a securely stored API key (server-side only, never exposed to the frontend). No custom model training occurs; Gemini is used purely as a language-understanding and natural-language-generation layer over structured business data retrieved from PostgreSQL.

9.2 Prompt Flow

User Question -> Intent Extraction (keyword/entity match) -> Data Retrieval (DB query) -> Prompt Construction (question + retrieved data + system instructions) -> Gemini API Call -> Natural Language Response -> Displayed to User

9.3 Response Flow

The backend constructs a prompt combining the user's question, a system instruction restricting the assistant to the provided business context, and the relevant retrieved data (e.g., low-stock product list, today's invoices). The Gemini response is returned as-is to the frontend chat window and additionally stored in the AILogs table for audit.

9.4 Business Query Handling

- Low stock queries retrieve products below threshold before prompting Gemini.
- Sales summary queries retrieve today's/period invoices and totals before prompting Gemini.
- Best customer queries retrieve top customers by spend before prompting Gemini.
- Inventory summary queries retrieve aggregate stock counts by category before prompting Gemini.

9.5 AI Limitations

- The assistant only answers questions related to the company's own business data; general-purpose queries unrelated to business operations are politely declined.
- The assistant does not perform any write/update actions on data — it is read-only and advisory.
- Response accuracy depends on the completeness of the underlying data and Gemini API availability.

10. Security Requirements

10.1 JWT Authentication

All authenticated API access uses short-lived JWT access tokens and longer-lived refresh tokens, following the flow defined in Section 3.1.

10.2 Role Based Access

Every protected endpoint enforces role checks server-side, as defined in Section 3.12, independent of any client-side UI restrictions.

10.3 Password Hashing

Passwords are never stored in plain text; Django's PBKDF2 (or a stronger configured hasher) is used for all stored credentials.

10.4 Session Timeout

Access tokens expire after 60 minutes of issuance; inactive sessions require re-authentication via refresh token or login after 7 days.

10.5 HTTPS

All traffic between the client, backend and the Gemini API occurs over HTTPS/TLS; the API key for Gemini is never exposed in frontend code.

10.6 Input Validation

All API inputs are validated server-side using Django REST Framework serializers, regardless of any frontend validation already performed.

10.7 Audit Logs

Authentication events (login, logout, failed attempts) and Admin actions (company activation/deactivation) are logged with timestamp and user identity for audit purposes.

11. Use Cases

Use Case ID	Name	Primary Actor	Description
UC-01	Register Company	Business Owner	Owner creates an account and registers a company with a selected business category.
UC-02	Login	Owner / Employee	User authenticates and is redirected to their category-specific dashboard.
UC-03	Add Product	Owner / Employee	User adds a new product to the Inventory module.
UC-04	Generate Invoice	Owner / Employee	User creates a sales invoice for a customer, updating stock automatically.
UC-05	Generate Report	Owner / Employee	User generates a Sales, Inventory or Customer report for a chosen date range.
UC-06	Chat with AI	Owner / Employee	User asks a natural-language business question and receives an AI-generated answer.
UC-07	Predict Sales	Owner	User views a next-day sales prediction generated by the Linear Regression model.
UC-08	Manage Customers	Owner / Employee	User adds, updates or reviews customer records and their purchase history.
UC-09	Manage Inventory	Owner / Employee	User searches, edits, and monitors stock levels of products.

12. Business Rules

The following business rules govern system behaviour across modules and are enforced primarily at the backend layer.

BR-001: A user account must have a unique, verified email address.

BR-002: A company must be linked to exactly one Business Category at all times.

BR-003: Business Category cannot be changed once transactional data exists for the company.

BR-004: Every business record (Product, Customer, Supplier, Employee, Invoice) must belong to exactly one Company.

BR-005: Only the Owner/Admin role may create or modify Employee accounts and their permissions.

BR-006: An Employee can only access modules explicitly granted by the Owner.

BR-007: Product stock quantity must never be negative.

BR-008: A product is flagged Low Stock automatically when its quantity falls below the configured threshold (default: 10 units).

BR-009: Deleted Products, Customers and Suppliers are soft-deleted, not permanently removed, if referenced elsewhere.

BR-010: An Invoice must contain at least one product line item.

BR-011: An Invoice quantity per line item cannot exceed the currently available stock of that product.

BR-012: Once confirmed, an Invoice cannot be edited; corrections require a new invoice.

BR-013: Confirming an Invoice automatically decrements the corresponding product stock quantities.

BR-014: Customer purchase history is updated automatically upon Invoice confirmation.

BR-015: Duplicate customers are identified by matching phone number within the same company.

BR-016: JWT access tokens expire 60 minutes after issuance.

BR-017: JWT refresh tokens expire 7 days after issuance.

BR-018: An account is locked for 15 minutes after 5 consecutive failed login attempts.

BR-019: Passwords must be at least 8 characters and include at least one number and one special character.

BR-020: A Notification must always reference a valid source record (Product or Invoice).

BR-021: Notifications are automatically archived after 30 days.

BR-022: Report data must always be scoped to the requesting user's own company.

BR-023: The AI Assistant may only access and send the requesting company's own data to the Gemini API.

BR-024: The AI Assistant is read-only and shall not perform any data modification actions.

BR-025: ML predictions must be computed using only the requesting company's own historical data.

BR-026: ML predictions must be clearly labelled as estimates, not guarantees.

BR-027: A minimum data threshold (e.g., 30 days of sales history) is required before Sales Prediction is enabled.

BR-028: K-Means Customer Segmentation always uses exactly three clusters, mapped to VIP, Regular and New.

BR-029: System Admin accounts are created only by direct database/administrative provisioning, not through public registration.

BR-030: Only the System Admin role can activate or deactivate a company account.

BR-031: A deactivated company's users cannot log in until the company is reactivated.

BR-032: All monetary values are stored and displayed in a single configured currency for the company.

BR-033: Email address cannot be changed after account registration.

BR-034: Company logo uploads are restricted to PNG/JPG formats under 2 MB.

BR-035: Every protected API endpoint must validate both authentication and role-based authorization.

BR-036: Audit logs for authentication and Admin actions must be retained for the duration of the academic evaluation period.

13. Assumptions

- Users have basic computer literacy and access to a modern web browser with stable internet connectivity.
- A valid Google Gemini API key with sufficient quota will be available throughout development and demonstration.
- Business data volumes remain at small-business scale suitable for a single PostgreSQL instance.
- The project will primarily be evaluated using representative seed/sample data rather than large-scale production data.
- Only one developer (the student) is responsible for implementation within the semester timeline.

14. Constraints

- The project must be completed by a single student within one academic semester.
- Only the specified technology stack shall be used: React.js, Tailwind CSS, Material UI, Python, Django, Django REST Framework, PostgreSQL, JWT, Scikit-Learn, Pandas, NumPy and the Google Gemini API.
- No Deep Learning, Neural Networks, or LSTM models are permitted.
- No Docker, Kubernetes, CI/CD pipelines or microservices architecture are required or expected.
- No native mobile application shall be developed.
- Voice AI, AI Agents, Plugin/Theme Marketplaces, Workflow Builder, Advanced Analytics, White Labelling, OCR, Blockchain and IoT are excluded from the current scope (see Section 16, Future Scope).

15. Acceptance Criteria

The project will be considered successfully complete when the following criteria are met:

- A user can register, log in, and register a company with a selected Business Category end-to-end without errors.
- The Dashboard and Sidebar correctly adapt to all four supported Business Categories.
- Inventory, Sales, Customer, Supplier and Employee modules support their defined CRUD operations correctly.
- Invoices correctly update stock and customer purchase history upon confirmation.
- Reports accurately reflect underlying transactional data for a selected date range.
- The AI Assistant correctly answers each of the five sample query types defined in Section 3.13 using real company data.
- Each of the three ML features (Sales Prediction, Customer Segmentation, Demand Prediction) returns a plausible, data-grounded output.
- Role-based access control correctly restricts Employee access according to assigned permissions.
- The application is responsive and functions correctly on desktop, tablet and mobile viewport widths.

16. Future Scope

The following features are explicitly out of scope for the current mini project but may be considered for future extension of AI-BOS beyond the academic submission:

- Voice AI — voice-based interaction with the Business Assistant.
- AI Agents — autonomous multi-step task execution beyond conversational Q&A.
- Plugin Marketplace — third-party extensions and integrations.
- Mobile Application — native iOS/Android apps for on-the-go business management.
- Advanced AI — deep learning-based forecasting and recommendation engines.
- Workflow Builder — no-code automation of business processes across modules.